

Solution Requirements

Date	10 July 2026
Team ID	XXXXXX
Project Name	Raising Waters
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>The system provides basic web-based access control for disaster management coordinators to securely log into the main monitoring panel.</i>	Medium
2	Authorization levels	<i>Disaster Coordinators have full access to input weather data, trigger alerts, and view analytics. Standard Users/Public can only view the active flood risk status dashboard</i>	High
3	External interfaces	<i>The system interfaces with external Weather APIs to pull real-time monsoon metrics, rainfall depth, and cloud coverage data.</i>	High

4	Transactions processing	<i>The system processes real-time weather data strings through the pre-processing module and feeds clean arrays directly into the ML inference pipeline.</i>	High
5	Reporting	<i>The application generates live visual flood risk alerts categorized into color-coded tier levels (e.g., Low, Medium, High Risk) on the main user dashboard.</i>	High
6	Business rules, etc.	<i>Flood alerts must be triggered instantly if the ML model evaluations (XGBoost/KNN) cross the designated risk thresholds based on cumulative rainfall metrics.</i>	High
7	Compliance to laws or regulations	<i>Data transmission and storage comply with regional open data standards for public environmental safety reporting.</i>	Medium
8	Other	<i>The system logs every prediction history instance into a dedicated datastore for model accuracy verification and future audits.</i>	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	The ML model must process weather inputs and deliver the risk tier output quickly.	<i>Inference latency under 2 seconds per request.</i>
2	Scalability	The system should seamlessly scale to handle an increased volume of incoming automated weather station data and concurrent user dashboard requests during peak monsoon seasons.	Supports processing up to 10,000 concurrent data stream requests without degradation
3	Security & Data Privacy	<i>The system must secure user login data and project credentials, ensuring communication between the Flask server and the database is protected</i>	<i>Implementation of standard HTTPS protocol and secure hashing for coordinator credentials.</i>
4	Reliability & Availability	<i>The system must remain consistently online to deliver real-time flood predictions and early warnings throughout emergency situations.</i>	<i>Minimum of 99.9% application uptime during active deployment months</i>
5	Usability & Accessibility	<i>The user interface of the Flask dashboard must be intuitive, clean, and optimized for quick scanning so disaster coordinators can make immediate decisions.</i>	<i>Dashboard page loads completely within 3 seconds on standard browser connections.</i>
6	Maintainability	<i>The project code and machine learning pipeline should be clearly documented to easily allow future updates to the predictive models or new weather feature parameters. Fully documented codebase with comments for easy model retraining scripts.</i>	Fully documented codebase with comments for easy model retraining scripts.